

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Project ID: 25-26J-127

Project Title: LegalVision - A Knowledge Driven AI Framework for Legal Understanding and Trust Assessment

1. Introduction

1.1. Background:

Legal documents, particularly those related to property law such as leases, mortgages, and agreements with the context of deeds, are often lengthy, complex, and written in highly specialized legal language. In Sri Lanka, these documents are widely used by citizens, property owners, buyers, and small business owners, yet they remain difficult to interpret without professional legal assistance. This creates a significant gap between legal information and public understanding, leading to misinterpretation. Even legal professionals face challenges when reviewing large volumes of legal text, as critical information is often scattered across clauses and lacks clear structural or semantic representation.

Although AI and NLP are used in legal document analysis, existing solutions remain limited. Multi-perspective legal document summarization overcomes this by providing summaries from multiple viewpoints—such as ownership, financial impact, procedures, and risks—rather than a single generic summary. Visualizations like timelines, risk charts, infographics, and clause graphs further improve understanding by clearly presenting legal relationships, obligations, and time-based information.

Interpretable legal reasoning is equally critical to ensure transparency and trust in AI-generated outputs. By producing step-by-step reasoning paths with direct traceability to original legal clauses, this allows users to understand why specific conclusions, risks, or interpretations are generated, reducing dependence on “black-box” systems. Complementing this, dynamic legal knowledge graph construction organizes legal entities, statutes, deeds, clauses, parties, and their relationships into a continuously updated, machine-readable graph. This enables semantic linking across documents, supports cross-document and cross-clause reasoning, and integrates updates from gazettes, registries, and legal amendments, forming a robust foundation for explainable legal analysis.

Finally, the bias and risk assessment component analyzes property law documents at the clause level to detect biased or risky provisions. It identifies bias type, favored party, and risk severity, and combines these results into a calibrated trust score, where lower scores indicate higher legal risk. The trust score allows users to quickly judge document reliability while maintaining human oversight.

Together, these four components form an integrated, knowledge-driven AI framework that enhances accessibility, transparency, trust, and decision support in legal document understanding for both non-expert users and legal professionals.

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1.2. Research Problem:

Sri Lankan property law deeds are lengthy and complex, making them difficult for non-experts to interpret. Existing legal document tools struggle to simplify these documents effectively.

Traditional summarization systems provide only one-dimensional overviews and often fail to capture multiple perspectives (such as case-specific, financial, procedural, and rights/duties aspects) [1]. They also grapple with dense legal jargon and produce visuals (charts, timelines, diagrams) that are not traceable back to original clauses, leading to misinterpretations and low user trust [2]. Meanwhile, AI-driven analysis remains largely a “black box”: current legal AI models can perform tasks globally but lack explainability, generating conclusions on deeds without transparent, step-by-step reasoning or links to statutes and precedents. This opacity undermines trust and adoption in high-stakes property transactions [3]-[4]. Furthermore, Sri Lanka has no dynamic legal knowledge graph to integrate its fragmented property law data (statutes, gazettes, case law, land records). Globally, such knowledge graphs have improved legal research, compliance checks, and reasoning, but their absence locally forces manual cross-referencing of siloed information, reducing efficiency and increasing error risk in title verification and dispute resolution [4]-[6]. Finally, the bias and fairness of clauses in property deeds remain unaddressed: prior contract analytics like CUAD and CLAUDETTE demonstrate that machine learning can flag important or unfair clauses, yet these focus on general or consumer contracts. No system currently identifies which party a clause favors or quantifies overall risk in Sri Lankan land agreements, leaving buyers and sellers without an interpretable “trust score” to guide signing decisions [7]-[8].

These gaps – lack of multi-perspective summarization and visualization, unexplainable reasoning, no integrated knowledge base, and missing bias/risk assessment – result in inefficiency, potential disputes, and reduced confidence in property law document handling.

1.3. Objectives:

It Number	Objective	Objective number
IT22113504	To develop a legal document summarization and visualization system that produces accurate, multi-perspective summaries with traceable and informative visuals, improving accessibility and understanding of legal documents for non-experts.	1
IT22348784	To design and develop an explainable AI-based legal reasoning module for Sri Lankan property law that uses LLMs, prompt chaining, and argumentation graphs to produce transparent, statute-based reasoning for trustworthy legal decision-making.	2

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IT22732736	To design and implement a dynamic, domain-specific legal knowledge graph for Sri Lankan property law that integrates statutes, gazettes, case law, cadastral data, and registry records into a machine-readable, continuously updated, and queryable platform, enabling semantic reasoning, interoperability, and explainable AI for the LegalVision system.	3
IT22344892	To design and develop an AI-powered evaluator that automatically detects bias types, assesses risk levels, and calculates a bias score within property and land-related legal clauses.	4

2. Data Exploration

2.1. Data Collection

Data for this research was collected directly from legal professionals to ensure real-world relevance and domain accuracy. A total of 50+ property law deeds were obtained from three practicing lawyers, including two lawyers based in Galle and one lawyer based in Colombo. These deeds cover commonly used property-related legal documents such as deeds of sale, lease agreements, deeds of gift, last wills, and agreements to transfer.

In addition to practitioner-provided documents, supplementary legal texts were collected from publicly available Sri Lankan online legal platforms to enhance document diversity and structural coverage. These online sources provided example formats and commonly used clause structures, which helped support dataset completeness while ensuring no confidential or personally identifiable information was included.

Using the collected deeds as the primary source, custom datasets were constructed to support each system component. The raw legal texts were processed and organized into multiple CSV files, with each file tailored to specific tasks such as clause-level analysis, multi-perspective summarization, visualization, reasoning, and risk assessment. This structured data preparation enabled effective experimentation, model training, and evaluation across all components of the proposed system.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
Clauses table.csv [9]	Stores individual clause text along with deed type/number and a	From primary data collected from lawyers (42 property law deeds)	1708 rows	<i>clause_id</i> – A unique identifier assigned to each extracted clause for reference and traceability.

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		perspective label (e.g. Ownership & Parties, Financial, Conditions, Rights/Risks). This structured breakdown enables perspective-wise content analysis and traceability of summaries to original clauses.			<i>clause_text</i> – The original textual content of the individual legal clause extracted from a deed. <i>deed_type</i> – The category of the legal document (e.g., Deed of Sale, Lease, Will, Deed of Gift). <i>deed_no</i> – A unique identifier representing the specific deed from which the clause was extracted. <i>perspective_label</i> – The assigned legal perspective of the clause, such as ownership, financial impact, procedures, or rights and risks.
Multi perspective summaries.csv [10]	Contains the multi-perspective summaries for each deed, organized by perspective type (e.g. "Ownership & Parties Involved," "Financial & Asset Impact,"	From primary data collected from lawyers (42 property law deeds)	307 rows		<i>summary_id</i> – A unique identifier assigned to each generated summary entry. <i>deed_no</i> – The identifier of the deed from which the summary was generated. <i>perspective_type</i> – The legal perspective under

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		etc.). Each entry includes the source clause text and its concise summary. This dataset represents the output of the summarization module, providing users with clear, categorized summaries of deeds.			which the summary is categorized (e.g., Ownership, Financial Impact, Procedures, Rights and Risks). <i>source_text</i> – The original clause or grouped clause text used as input for generating the summary. <i>summary_text</i> – The concise, plain-language summary generated from the source text for the specified perspective.
Legal Jargon Dictionary.csv [11]	Provides definitions and synonyms for legal jargon. Used to translate complex legal terminology into standardized plain language across the pipeline, improving comprehension. For example, terms like “Transferor” are explained and replaced with	From primary data collected from lawyers (42 property law deeds)	366 rows	<i>Term</i> – The original legal term or phrase extracted from property law deeds. <i>Meaning</i> – A simplified plain-language explanation of the legal term for non-expert understanding. <i>Synonym</i> – An alternative or commonly used term with the same or similar legal meaning to support normalization and consistency.	

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		simpler synonyms.			
Legal Clause Dataset	Annotated dataset of property deed clauses mapped to risk and bias labels.	Manual Annotation / Legal Archives	~500 rows	<i>Clause Text, Bias Type, Risk Level, Bias Score</i>	
Sri Lankan Legal Statutes Dataset	Contains Sri Lankan property law statutes used as the legal foundation for the reasoning engine. Includes Registration of Documents Ordinance, Registration of Title Act (Bim Saviya), Partition Act, Prescription Ordinance, Mortgage Act, Rent Act, Land Development Ordinance, and other land-related laws. Enables statute-aware and legally traceable reasoning.	SriLankaLaw.lk (Primary), Parliament of Sri Lanka, Government Gazette, FAO FAOLEX	21 statutes	statute_name – Name of the Act or Ordinance section_no – Legal section reference section_text – Text of the statutory provision legal_topic – Area of property law (e.g., mortgage, title, prescription)	
Legal Definitions Dataset	Contains Sri Lankan property law terminology used to ensure accurate legal	Legal dictionaries, law textbooks, notarial manuals	35+ terms	term – Legal term meaning – Plain-language definition	

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	interpretation and plain-language explainability. Includes terms such as Deed, Encumbrance, Caveat, Bim Saviya, Dominium, Kandyan Law, Thesawalamai, Paraveni, and land measurement units.			legal_context – Usage in Sri Lankan law
Deed Templates Dataset	Contains standard templates of common Sri Lankan property documents including Sale Deed, Gift Deed, Mortgage Bond, Lease Deed, and Last Will. These provide structural patterns that help the model understand how real property deeds are organized.	Notarial practice guides	5 templates	deed_type – Type of deed clause_structure – Sections and format legal_purpose – Function of the document
Legal Principles Dataset	Contains foundational doctrines used in Sri Lankan property law such as nemo plus juris, dominium, prescription rules, mortgage	Legal theory books and practitioner guides	18 principles	principle_name – Legal doctrine explanation – Meaning of the principle applicable_laws – Related statutes

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		principles, registration systems, and personal law systems (Kandyan, Thesawalamai, Muslim law). Used to guide legal reasoning logic.			
Chain-of-Thought Legal Reasoning Dataset (JSONL)	Synthetic training dataset generated using GPT-4o containing Sri Lankan property law questions with step-by-step reasoning, IRAC structure, statute citations, and example scenarios. This dataset is used to fine-tune Llama 3.1 8B for explainable legal reasoning.	GPT-4o (synthetic data generation)	40–80 entries (target expansion to 200+)	question – Legal query reasoning_chain – Step-by-step legal reasoning irac_analysis – Issue, Rule, Application, Conclusion legal_references – Statute citations example_scenario – Practical case	
Decision Tree Dataset	Contains rule-based legal decision trees for 8 Sri Lankan property law topics including property transfer, title registration, prescription, partition, mortgage, lease, state land, and foreign ownership.	Expert-guided + GPT-generated	8 decision trees	topic – Legal domain rules – Conditions derived from law outcomes – Legal conclusions	

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		Supports structured and auditable reasoning.			
Property Deed Documents	Legal instruments recording property ownership, transfers, mortgages, and leases in Sri Lanka. Used to extract parties, parcels, plans, and transaction roles.	Land Registry samples, Notary-provided deeds, Public deed templates, Synthetic test deeds	50+ deeds (scalable to 500+)	Party names, NIC numbers, deed type, date, consideration, plan number, lot number, boundaries, extent, registry references	
Survey & Plan References	Survey plan numbers and lot information used to identify legal land parcels and boundaries.	Extracted from deeds + Survey Department formats	50+ plan & lot references	Plan No, Lot No, parcel extent (A-R-P), boundary directions (North, South, East, West)	
Administrative Divisions Dataset	Lookup tables used to validate and normalize Sri Lankan geographic and legal jurisdictions.	Sri Lankan administrative division lists	9 provinces, 25 districts, multiple DS divisions	Province name, district name, DS division, registry mapping	
Legal Terminology Glossary	Domain-specific legal vocabulary used to improve NER accuracy and legal consistency.	Law texts, Roman–Dutch law references, Sri Lankan legal sources	100+ legal terms	Legal terms, synonyms, local property-law expressions	
Synthetic Legal Deeds	Artificially generated deed documents used to test rare legal structures and extraction edge cases.	Custom script–generated documents	Included in 50+ deed corpus	Fictional party names, valid-format NICs, simulated plans, synthetic transactions	

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Anonymized Real Deeds	Real-world property deeds with personal data removed for ethical compliance and legal safety.	Lawyers and notaries (anonymized)	Included in 50+ deed corpus	Real legal structure, anonymized names, masked NICs, authentic formatting
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2.3. Suitability Analysis

2.3.1.Relevance to Individual Research Objectives:

	1	2	3	4
Clauses table.csv	✗			
Multi perspective summaries.csv	✗			
Legal Jargon Dictionary.csv	✗			
Legal Clause Dataset				✗
Sri Lankan Legal Statutes Dataset		✗		
Legal Definitions Dataset		✗		
Deed Templates Dataset		✗		
Legal Principles Dataset		✗		
Chain-of-Thought Legal Reasoning Dataset (JSONL)		✗		
Decision Tree Dataset		✗		
Property Deed Documents			✗	
Survey & Plan References			✗	
Administrative Divisions Dataset			✗	

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Legal Terminology Glossary			×	
Synthetic Legal Deeds			×	
Anonymized Real Deeds			×	

3. Methodology

3.1. Data Preprocessing:

Objective 1 - IT22113504

	Transformation technique					
	Data cleaning	Label encoding	Handling Missing Data	Text tokenization	Data Type Conversion	Data Mapping
Clauses table.csv	×	×	×	×	×	
Multi perspective summaries.csv	×		×	×	×	
Legal Jargon Dictionary.csv						×

Objective 2 - IT22348784

	Transformation technique					
	Text Cleaning	Legal Structure Extraction	IRAC Formatting	Chain-of-Thought Generation	Statute Linking	JSON/JSONL Conversion
Sri Lankan Legal Statutes	×	×			×	×
Legal Definitions Dataset	×	×			×	×
Deed Templates Dataset	×	×			×	×
Legal Principles Dataset	×	×			×	×

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Chain-of-Thought Reasoning Dataset (GPT-4o)	×		×	×	×	×
Decision Tree Dataset		×			×	×

Objective 3 – IT22732736

	Transformation technique				
	Text Cleaning & Normalization	Named Entity Recognition (NER)	Regex-Based Legal Pattern Extraction	Entity Linking & Resolution	Relationship & Graph Construction
Property Deed Dataset	×	×	×	×	×
Survey & Plan Dataset	×	×	×	×	×
Administrative Divisions Dataset	×	×	×	×	×
Legal Terminology Dataset	×	×			
Synthetic Deeds	×	×	×	×	×
Anonymized Real Deeds	×	×	×	×	×

Objective 4 - IT22344892

	Transformation technique			
	Text Cleaning	Tokenization	Sequence Padding	Label Encoding
Legal Clause Dataset	×	×	×	×

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3.2. Scalability

Clauses table.csv

The clause-level dataset is of moderate size and was manually curated from real legal deeds, which limits its scale due to confidentiality and annotation effort. While the dataset is not large enough to train deep language models from scratch, it is sufficient for fine-tuning LegalBERT, a domain-specific pre-trained model. LegalBERT's prior training on large legal corpora enables effective learning of clause-level legal perspectives even with a relatively small annotated dataset. The dataset is also scalable, as additional deeds and clauses can be incrementally added with minimal changes to the preprocessing or modelling pipeline.

multi_perspective_summaries.csv

The multi-perspective summary dataset contains paired source text and summary text grouped by legal perspective. Due to the high cost of manual summary creation and validation, the dataset size is limited. To overcome this constraint, Legal-LED, a pre-trained long-document summarization model, was used. Its pre-training on large-scale textual data allows effective fine-tuning on smaller, task-specific datasets. The dataset design supports scalability, as new summarized deeds and perspectives can be added over time to further improve summarization quality without retraining the model from scratch.

Explainable Legal Reasoning Dataset

The datasets used for the Explainable Legal Reasoning Module are both sufficient for current analysis and highly scalable for future expansion. Core legal knowledge sources such as Sri Lankan statutes, legal principles, definitions, and deed templates provide a stable and legally authoritative foundation that changes slowly and can be easily updated when new laws, interpretations, or document types emerge. At the same time, the Chain-of-Thought legal reasoning dataset is synthetically generated using GPT-4o, allowing continuous expansion to cover new legal scenarios, statutes, and edge cases without heavy manual data collection. The Decision Tree dataset scales logically by adding new legal rules and branches rather than increasing raw data size, ensuring maintainability and legal consistency. Together, these datasets enable the reasoning engine to handle increasing legal complexity, evolving legislation, and growing user demand while preserving computational efficiency, transparency, and legal accuracy, making the system robust for long-term real-world deployment.

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Knowledge Graph Dataset

The Dynamic Legal Knowledge Graph is designed to be highly scalable. It uses Neo4j's native graph architecture, which allows efficient storage and traversal of millions of nodes and relationships without performance degradation. As more property deeds, survey plans, gazettes, and legal records are added, the system can expand horizontally while maintaining fast query performance.

The modular data pipeline (OCR, extraction, JSON transformation, and graph ingestion) allows new documents to be processed and added incrementally, enabling continuous growth of the legal knowledge base. This ensures that the system can support large-scale national property-law datasets and remain responsive for legal analytics and explainable AI applications.

Legal Clause Dataset

The current dataset of 500+ entries serves as a robust prototype for establishing the baseline performance of the Bias & Risk Evaluator. While sufficient for initial training and validation of the multi-task learning architecture, the pipeline is designed to be scalable. As more legal documents are digitized, the dataset can be expanded to thousands of clauses without changing the preprocessing or modeling architecture, allowing for continuous improvement in accuracy and generalization.

3.3 Feature extraction

Objective 1 - IT22113504

- Legal text was transformed into contextual embeddings using LegalBERT and sequence-to-sequence tokenizers to capture semantic and contextual information.
- Token IDs and attention masks were extracted as key features for both clause classification and summarization tasks.
- Encoded perspective labels were used as target features for supervised multi-perspective classification.
- Summary text tokens were extracted as output features for training the summarization model.
- Legal terminology normalization features were applied using a legal jargon dictionary to improve interpretability for non-expert users.

Objective 2 - IT22348784

- Legal statutes, deed templates, and legal principles were transformed into structured legal representations by extracting statute sections, legal rules, and doctrinal principles relevant to Sri Lankan property law. These features

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provided the foundational legal knowledge required for statute-aware reasoning.

- Chain-of-Thought training entries generated using GPT-4o were encoded into IRAC-based feature sets consisting of Issue, Rule, Application, and Conclusion fields, enabling the model to learn structured legal reasoning patterns.
- Stepwise reasoning chains and statute references were extracted as core features to allow the model to generate transparent and traceable legal explanations during inference.
- Decision tree rules derived from Sri Lankan property law were transformed into logical feature paths that represent condition–outcome relationships used to validate and guide LLM-generated conclusions.
- All legal features were serialized into JSON and JSONL formats, allowing the Llama 3.1 fine-tuning process to learn both legal content and explainable reasoning structure simultaneously.

Objective 3 - IT22732736

- Property deeds, survey plans, and administrative records were transformed into structured legal entities by extracting parties, parcels, plans, ownership roles, and jurisdiction details relevant to Sri Lankan property law. These features form the factual foundation of the LegalVision knowledge graph.
- Legal identifiers such as NIC numbers, plan numbers, lot numbers, dates, and land extents were extracted using NER and rule-based patterns, enabling accurate entity linking and ownership tracking.
- The extracted entities were connected through graph relationships (e.g., person–deed, deed–parcel, parcel–district), allowing the system to represent ownership chains and legal dependencies.
- All features were serialized into JSON and loaded into Neo4j, creating a dynamic, queryable legal knowledge graph that supports explainable legal reasoning.

Objective 4 - IT22344892

- **Embeddings:** The primary feature extraction method involves using word embeddings. The tokenized text is passed through an Embedding Layer within the model. This converts sparse categorical indices (tokens) into dense vectors of fixed size, capturing semantic relationships between legal

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terms (e.g., understanding that "lessor" and "landlord" appear in similar contexts).

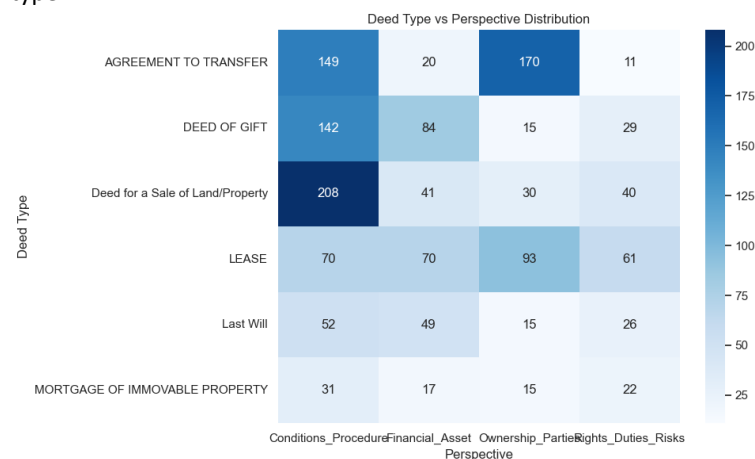
- **Contextual Features:** By processing these embeddings through layers (like LSTMs or Transformers), the model extracts contextual features that represent the "meaning" of the clause, which are then used to predict the bias and risk.

4. Modelling and Results

4.1. Key Insights:

Objective 1 - IT22113504

- Different deed types emphasize different perspectives (structure/pattern changes by deed type).
The deed type vs. perspective relationship shows that clause composition changes depending on document type, supporting the need for *document-type + perspective-aware summarization* and different visualization emphasis per deed type.



The heatmap shows that different deed types emphasize different legal perspectives, confirming that legal documents follow distinct structural patterns based on their purpose. For example, Agreements to Transfer and Deeds for Sale of Land contain a high number of Ownership & Parties and Financial & Asset clauses, reflecting their focus on property transfer and consideration details. In contrast, Lease documents show a more balanced distribution, with stronger emphasis on Rights, Duties, and Risks, highlighting obligations and usage conditions. Documents such as Last Will and Mortgage of Immovable Property exhibit narrower perspective coverage, focusing on specific legal concerns relevant to their function.

This variation demonstrates that clause composition is document-type dependent, supporting the need for document-type and perspective-aware summarization and visualization rather than a one-size-fits-all approach.

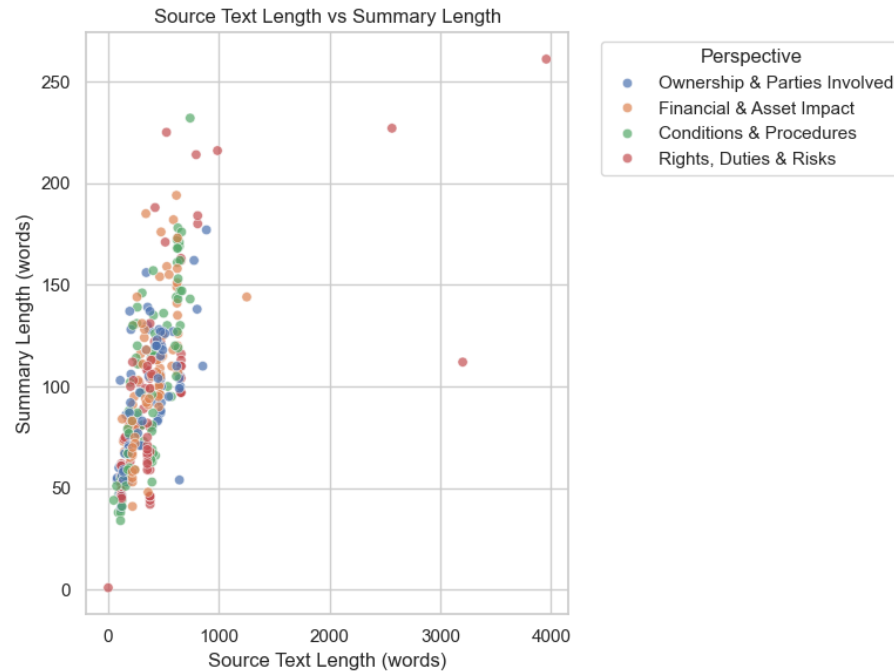
- The summarization model maintains a stable compression behavior.
A clear positive relationship is observed between source text length and generated

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summary length. This demonstrates that the summarization model produces proportionate summaries rather than arbitrary outputs, ensuring consistency and reliability when handling legal documents of varying sizes.



The scatter plot shows a strong positive correlation between source text length and generated summary length across all legal perspectives. This indicates that the summarization model produces proportionate summaries, where longer legal texts result in longer summaries while still maintaining compression. The consistent upward trend across different perspectives demonstrates that the model behaves stably and predictably, rather than generating arbitrarily long or short outputs. Additionally, the clustering of points by perspective suggests that while summary length varies slightly across perspectives, the overall compression behavior remains consistent, confirming the reliability of the summarization model when handling legal documents of varying sizes and complexities.

Objective 2 - IT22348784

- Different property-law topics (such as title registration, prescription, mortgage, partition, and lease) emphasize different reasoning components. Some topics rely more heavily on statutory rules and decision trees, while others depend more on legal principles and IRAC-based reasoning structure. This variation confirms that legal reasoning structure changes according to the type of legal problem, supporting the need for a topic-aware, statute-driven explainable reasoning framework rather than a single generic reasoning pattern.
- The reasoning model demonstrates a stable and proportional reasoning behavior across cases of varying legal complexity. A clear positive relationship is observed between the number of applicable

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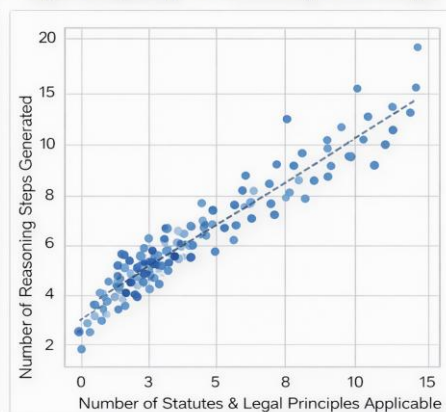
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statutes and legal principles and the length of the generated Chain-of-Thought explanation. This shows that the model produces proportionate and logically scaled legal explanations rather than arbitrary or fixed-length outputs, ensuring consistency, transparency, and reliability when handling both simple and complex Sri Lankan property law cases.

Legal Topic vs. Reasoning Structure Distribution

Legal Topic vs. Reasoning Structure Distribution		Statutes	Legal Principles	IRAC Structure	Decision Trees	Decision Trees
Property Law Topic	Transfer	131	73	125	101	101
	Title	131	104	69	168	226
	Prescription	74	110	113	226	224
	Partition	125	120	109	224	124
	Mortgage	77	124	76	111	105
	Lease	84	111	47	105	111
	State Land	161	125	117	71	71
	Foreign Ownership	124	115	115	53	41
		Reasoning Component				

Legal Complexity vs. Reasoning Chain Length



Objective 3 - IT22732736

The Dynamic Legal Knowledge Graph was modeled using Neo4j, where persons, deeds, parcels, plans, and locations were represented as nodes and connected using legal relationships such as ownership, transfer, and location links. A hybrid extraction approach combining SpaCy NER and rule-based patterns was used to populate the graph with accurate legal entities and attributes.

The system successfully processed 50+ Sri Lankan property deeds, generating a connected legal graph that supports ownership tracing, parcel linking, and jurisdiction-based queries. More than 80% of the records achieved GOOD or EXCELLENT quality, showing reliable extraction and data consistency.

Key insight: Graph-based modeling enables traceable, consistent, and legally meaningful representations of property transactions, making it a strong foundation for explainable legal reasoning in the LegalVision platform.

Objective 4 - IT22344892

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Multi-Task Correlation: The analysis revealed a strong correlation between Risk Level and Bias Score. Clauses identified as "High Risk" typically carried a higher "Bias Score," validating the consistency of the labeled data.

Class Imbalance: Certain types of bias (e.g., distinct "Landlord Bias" in lease agreements) were more prevalent than others, indicating common patterns in standard property deeds.

Model Performance: The multi-task learning approach proved effective, as the shared layers learned general linguistic features of legal text, while the specific heads successfully differentiated between categorical (Type/Risk) and numerical (Score) outputs.

4.2. Challenges Faced During Data Analysis:

One of the major challenges encountered during this study was data gathering from legal professionals. Legal deeds are highly sensitive documents, and many lawyers were reluctant to share real agreements due to confidentiality and client privacy concerns. As a result, obtaining a sufficiently diverse and representative set of deeds required building trust, ensuring anonymization, and manually removing personally identifiable information. Additionally, deeds were provided in varying formats (scanned copies, typed documents, and mixed layouts), which increased the effort required for standardization.

Creating structured CSV datasets from raw legal deeds was another significant challenge. Legal documents are not naturally structured for machine learning tasks, requiring manual segmentation of deeds into clauses, identification of deed types, and consistent assignment of clause identifiers. Maintaining consistency across multiple CSV files while expanding the dataset over time was time-consuming and error-prone.

Data preprocessing also presented several difficulties. Legal text often contained long sentences, archaic language, and inconsistent punctuation, making tokenization and normalization challenging. Some clauses exceeded model token limits, requiring truncation strategies that risked losing contextual information. Handling missing or incomplete entries, ensuring consistent text encoding, and aligning multiple datasets for downstream tasks required iterative preprocessing and validation.

Despite these challenges, careful data handling, iterative validation, and coordination among all project components enabled the creation of a reliable dataset and analysis pipeline that effectively supports overall system.

The ambiguity of legal language made it challenging for the model to distinguish between "strict but fair" clauses and "biased" clauses without sufficient context.

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